

The UK Linguistics Olympiad 2017



Solution and marking.

Scoring: 1 point for each correct word. (Max 40)

- When two translations are possible, only credit unrepeated words.

a	masita / masāga kāpī cha.	4
b	Bindusita / Bindusāga ek baṭṭā churoṭ chaina.	6
c	Gitako śrīman cha.	3
d	Rameshki nātini cha.	3
e	timīsāga / timīsita das rupiy* chan? (*= a mixture of ã and ā)	5
f	Gharka tīnvatā koṭṭaharu chan.	4
g	parivārki nātini chaina.	3
h	timīsāga / timīsita miṭhāī chan?	4
i	Anilsita / Anilsāga s*co chaina. (*= a mixture of ã and ā)	4
j	masita / masāga pēnsila chaina.	4

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Commentary

Possession is expressed in Nepali with the possessed as subject, as though we said ‘A bike belongs to me’ instead of ‘I have a bike’. More precisely, the verb used is ‘be’, so a literal translation into English would be ‘A bike is (to) me’, where ‘to’ corresponds to the ending on the possessor noun, explained below. These suffixes are added to the possessor, which in these examples is either a name or a pronoun.

Suffix	explanation
-sita	Can be used interchangeably with – sāga. Both are used when the possessed is a small item that a speaker / person can carry with them, on their person, such as money, pens, cigarettes etc.
-sāga	See above.
-ko	Used when the possessed is too big to be carried by a person (such as houses or cars) or to refer to friends / family “belonging” to a person. The “ko” suffix is used when the possessed is masculine.
-ki	Used when the possession is too big to be carried by a person (such as houses or cars) or to refer to friends / family “belonging” to a person. The “ki” suffix is used when the possessed is feminine.
-ka	Used when the possession is too big to be carried by a person (such as houses or cars) or to refer to friends / family “belonging” to a person. The “ka” suffix is used for nouns (in possession) that are plural.

Your name:

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Solution and marking.

Scoring (max 30)

- 2.1 (max 6)
 - 2.1.a: 2 points (1 point with one error)
 - 2.1.b,c: 2 points (1 point with one incorrect morpheme)
- 2.2 (max 24)
- one for each position statement
- one for each meaning; ½ for a partial meaning
- two for a good explanation of conventions

a. kewa:pamiehmena:n	You (singular) see us.	2
b. We see them.	ne wa:pam a: ehm ena:n a ki	2
c. They see me.	ne wa:pam ekw ehm aki	2

2.

affix	position	meaning
ke (or k)	-1/prefix	2/second person/you
ne (or n)	-1/prefix	3/third person/he-she-they/him-her-them
eθ	+1/first suffix	1-2 / 1 subject, 2 object/ etc.
a:	+1/first suffix	1-3
i	+1/first suffix	2-1
ekw	+1/first suffix	3-1
ehm	+2/second suffix	? unclear. Maybe present tense?
wa:	+3/third suffix	plural 2
ena:n	+4/fourth suffix	plural 1
a	+5/fifth suffix	3/third person/etc.
ki	+6/sixth suffix	plural 3

Conventions:

- -n/+n: position
- 1, 2, 3: first, second, third person
- 1-2, etc: subject = first person, object = second person

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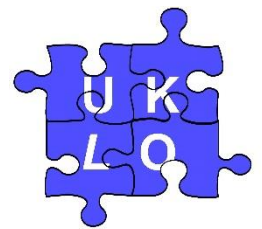
Solution and marking.

Scoring: 1 point for each correct cell (max 20)

name	1	2	3	4	5	6	7	8	9	10
description	D	I	O	F	G	B	Q	A	C	L
name	11	12	13	14	15	16	17	18	19	20
description	E	P	K	H	R	M	S	N	J	T

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Solution and marking.

Scoring: one point for each correct cell (max 16)

1	2	3	4	5	6	7	8
M	J	I	E	F	C	K	---
9	10	11	12	13	14	15	16
L	N	D	G	A	H	P	O

Commentary

This problem illustrates the inefficiency of the hieroglyph system, in that it requires a lot of guesswork because there's so little that's systematic. However, there are clues which will take you into the data.

- The last example ('cows') in the first table (in the introduction) suggested that plurality may be shown by three vertical lines. In that example, the lines are in a column, but three examples in the problem data have three vertical lines in a row: |||; since layout in general seems very variable, it's reasonable to assume that these three lines also mark plurality. Three of the hieroglyph patterns (2, 15, 16) include three lines, and three of the translations are plural (gifts, offerings, divine offerings). Hieroglyph patterns 15 and 16 share two glyphs, so let's assume that these are 'offerings' or 'divine offerings'.
- Of these two, 15 starts with a little flag which is repeated, again at the start, in two other patterns: 12 and 14. Maybe this means 'divine', i.e. to do with religion? If so, **15 = P** and **16 = O**. The other two patterns could mean 'priest' or 'priestess' (G, H). These contain glyphs representing people who could be men and women, and at a guess G looks more like a man and H like a woman. So **12 = G** and **14 = H**.
- Pattern 12 is the same as 11, apart from the 'divine' flag, so maybe 11 means 'male slave' and 12 means 'male slave of the gods' (i.e. 'male priest'). So **11 = D**. Similarly, 14 is very similar to 13, except for the divine flag, though here 13 has a single vertical line (like the ones in 11 and 12) which is not 14. So **13 = A**.
- Now that we have the glyph for 'woman', we can identify the only other pattern that contains it, 10, as 'woman'. So **10 = N**. We also notice that this glyph occurs with the little half-circle glyph in 10, 13 and 14, which was also found in the pattern for /nfrt/, 'cows' (but not in /nfr/); and that /hmt/, meaning either 'woman' (N) or 'female slave' (A), contains /t/. So maybe the half-circle glyph means /t/ and/or 'female'? The phonetic interpretation fits with the two t's matched by two half-circles in /htpt/, 'offerings' (O=16).
- One way forward now is to develop the idea that the little half-circle is /t/ and to apply this to two glyph patterns that are distinguished only by the half-circle: 7 and 3. If we look through the translations, we find a pair of pronunciations distinguished only by /t/: /pr/ 'house' (C) and /prt/ 'to go out' (K); and /prt/ 'procession' (I) and /prt/ 'seed' (J). However, our two glyph

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patterns (3 and 7) also include the walking-legs glyph which makes good sense with 'to go out' (K) and 'procession' (L). So 7 = K and 3 = L.

- Most of the remaining glyph patterns contain the rectangle glyph with a gap: maybe a simple ground plan of a house with a door? In that case, one of them presumably means 'house' (C)? The simplest patterns are 5 and 6, but the single straight line is simpler than the club-like glyph, so we can guess that the two-part pattern in 6, consisting of a house-glyph and a straight line (maybe meaning 'one item', as in 11 and 12), means 'house'. So 6 = C.
- The other pattern with the club-like glyph (5) is also used as part of a larger one (4), which contains a second house-glyph. Among the remaining words, we see a similar relation between 'pharaoh' and 'palace' (meaning 'pharaoh's house'). So 5 = F and 4 = E.
- This leaves just four patterns to identify. 1 and 2 are clearly related because both share three glyphs:
 - the house-glyph found in /pr/ 'house'
 - a mouth-shaped glyph
 - the half-circle glyph found in /hmt/, /prt/ and so on, which may indicate /t/.
- Given this similarity, we can assume that they correspond to meanings J (/prt/ 'seed') and M (/prt/, 'winter-season'). If so, we might guess that the mouth-shaped glyph can mean /r/, with /p/ for the house-glyph and /t/ for the half circle. To decide how to pair the patterns and words, we notice that 2 contains the three vertical lines of plurality, indicating countability or multiplicity; and it also contains a plough. So maybe 2 means 'seed' and 1 means 'winter-season'? If so, 1 = M and 2 = J.
- This leaves just two patterns: 8 and 9, and two words: B (/nw/ 'gifts') and L (/hms/, 'to sit down'). We know that one pattern and one word will remain unassigned, so we need to decide which of pattern is assigned to a word. The evidence supports 9 = L:
 - The little man seems to be sitting, so in this case the relevant fact is that he's sitting rather than that he's a man.
 - The first glyph (reading from the top) in 9 is the same as the first glyph in 10, /hmt/ 'woman'. So maybe this glyph means /h/, or even /hm/, as in L: /hms/.
 - We know from the very first example (in the introduction) that /w/ is indicated by a little bird. (At least, this is the most likely interpretation because the word concerned, /yw/ 'is', has so little meaning.) So if one of the remaining patterns did spell /nw/ 'gifts' (B), we might expect a little bird; but neither of the remaining patterns contains one, so they probably don't represent word B.
- So pattern 8 and word B are unassigned.

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Solution and marking.

Scoring (max 31)

- 5.1: 2 points for every correctly-filled cell. (max 16)
- 5.2: 1 point for every correct Yup'ik word. (max 5)
- 5.3: (max 10)
 - 3 points for a mathematical explanation of the magic square.
 - 5 points for a linguistic explanation of the Yup'ik number system.
 - 2 points for an explanation of how the clue numbers lead to the solution.

5.1

	1	2	3
a	4	9	2
b	3	5	7
c	8	1	6

5.2 Yuinaat yuinaq malruk akimiaq atauciq.

5.3 Explanation: see the commentary.

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Commentary (by the problem's author, Kai Low Rui Hao)

Candidates should recognize this as a mathematics problem and take into consideration the possibility of the spelling of numbers being represented in a different base (for example, in French, 99 is *quatre-vingt dix neuf*, which translates to $4 \times 20 + 10 + 9$).

There is only one possible type of 3-by-3 magic square, although there are 8 distinct configurations that stems from the one type due to rotation and reflection. These permutations are avoided due to the presence of the Yup'ik cross-number puzzle clues.

Since there are 9 digits, the numbers can be arranged via this formula

N-1	N+4	N-3
N-2	N	N+2
N+3	N-4	N+1

This formula takes into consideration 9 consecutive numbers, as well as the fact that the sum of every row, columns, and diagonals is the same – $3n$. Since the sum of each column is $3n$, the sum of three columns is $9n$. The sum from 1 to 9 is 45. Hence $9n = 45$, leaving us with $n=5$.

The answer to Task 1 is as above. Candidates may in the process arrive at the other 7 possible combinations. These permutations will be eliminated when checked against the yup'ik hints.

Since the question provided 2Down as a 3-digit number that starts with 9, and all numbers in the clues are 3-digit, and all Yup'ik numbers begin with Yuinaat, candidates have to consider that Yup'ik may be using a base larger than 10. The most common, in fact, is the Vigesimal system, which is base 20. This is also practical based on the background provided in the question – Yup'ik people based their concept of counting on body parts (20 fingers + toes).

Despite the various permutations, candidates should be able to arrive at 2D, which is 951. Further attempts at solving the spelling will reveal that $951 = (20 \times 20 \times 2) + (20 \times 7) + (10 + 1)$. This is true to the base 20 system. Arriving at this conclusion will reveal:

1. Suffix -k: multiply by 2/double/to do with two (a dual number)
2. Suffix -t: multiply by more than 2 (a plural number)
3. Suffix -q: the root suffix.

If the word ends in suffix -q, it signifies to the candidate that the subsequent number should be an addition and not a multiplication.

Candidates can then work on 3Across, which is the next biggest number with $20 \times 20 \times 2$. Hence, the number, which is the second biggest after one starting with "9", should start with "8". This will lead to 3A being 816. Candidates can then associate *akimiaq atauciq* to 16. From the 2D, *atauciq* = 1, hence *akimiaq* is 15. This is a reasonable and valid guess since Yup'ik people pay attentions to numbers based on the hands and feet.

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Next, refer to 3D where candidates can make out $(20 \times (10+?)) + 16$. Given that diagonals also sum up to 15, we can gather 2 and then 7 from it being the remaining cell in 3D. Hence 3D is 276 and it is $(20 \times (10+3)) + 16$.

Knowing that *qula* = 10, *akimiq* = 15, and that *malruk* = 2, *pingayun* = 3, we will be able to solve most of the magic square. The other number not mentioned is *cetaman* = 4.

Task 2: Yuinaat yuinaq malruk akimiq atauciq.

The diagonal 1a-2b-3c is 456 and can be expressed in Yup'ik spelling as $(20 \times 20) + (20 \times 2) + 16$. However, notice from other clues that numbers below 800 are not phrased in this manner but more of $(20 \times 22) + 16$. Hence the number will be expressed as $(20 \times (20 + 2)) + 16$.